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Endurance and electromyographic assessment of abdominopelvic motor control in women with primary dysmenorrhea: a cross-sectional study

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Primary dysmenorrhea (PD), a prevalent menstrual condition characterized by pelvic pain during the menstrual cycle, significantly impacts the quality of life of women and produces increased pain sensitivity that can persist throughout the menstrual cycle. However, scientific literature has not studied whether there are implications for alterations in muscle function and endurance in the abdominopelvic region during the non-painful phases of the menstrual cycle. The aim of this study was to compare muscle function and endurance capacity in the abdominopelvic region in women with PD versus women without this condition. An observational, cross-sectional study was designed to analyze muscle activation and endurance capacity using electromyography (EMG) during McGill exercises. Forty-four women were included, 22 with PD and 22 without dysmenorrhea. The results did not indicate significant differences in muscle activation and endurance of the abdominopelvic musculature between the two groups ($p > 0.05$). However, the analysis suggests that women with primary dysmenorrhea might develop compensatory strategies that allow them to maintain physical function despite their condition. These results suggest that the approach to PD could focus more on pain management rather than physical functionality, and more studies are needed from a comprehensive approach to more accurately evaluate the relationship between PD and muscle function.

Keywords Primary dysmenorrhea, Muscle function, Muscle endurance, Muscle activation, Electromyography, Pain menstrual cycle

Primary Dysmenorrhea (PD) is a common menstrual condition characterized by cramping abdominal and lumbopelvic pain during the menstrual cycle, often accompanied by general discomfort, and even nausea and vomiting. In some cases, these symptoms do not easily subside with pharmacological treatment, thus significantly affecting the quality of life of women suffering from it[1,2]. In addition to the acute pain experienced during menstruation, dysmenorrhea can involve a range of alterations in muscle function and endurance capacity in the abdominopelvic region[3]. PD occurs in 45–90% of women between menarche and menopause, and its diagnosis involves excluding pelviperineal, lumbopelvic, or gynecological pathologies that could explain the accompanying symptoms[3,4].

Research has shown that women with PD exhibit heightened pain sensitivity even during phases of the menstrual cycle without pain, suggesting an alteration in pain processing that extends beyond menstruation[2,3]. This alteration may stem from a process known as central sensitization[5], in which the central nervous system abnormally amplifies pain perception, even without tissue injury or inflammation[6]. Central sensitization has been associated with a greater risk of developing chronic pain conditions, indicating that PD could have more enduring impacts on somatosensory function[7]. However, specific studies on how PD affects muscle function and motor control in the abdominopelvic region remain limited, leaving a research gap regarding the potential impact of PD on muscular stability and motor control across the menstrual cycle. Existing scientific literature has

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indicated a relationship between dysmenorrhea and structural as well as functional changes in the musculature of the abdominopelvic region, as observed through different imaging tests[8,9]. This decrease in muscle thickness or cross-sectional area (CSA) could result in a reduction in muscle strength[10]. These findings support the hypothesis that dysmenorrhea can have a significant impact on the muscular function of the abdominopelvic region, highlighting the importance of addressing not only the pain but also the muscular alterations that may occur in these patients.

Moreover, research on motor control in the context of chronic pain suggests that changes in muscle composition may alter recruitment and motor control patterns[11,12]. Specifically, an abnormality in the order and amount of activation of superficial muscles compared to deep muscles has been detected in individuals experiencing pain[12,13]. In women with PD these changes have even been shown to affect postural stability[14]. However, research specifically investigating the motor control alterations in women with PD remains limited. An in-depth examination of these control alterations in this demographic could facilitate the implementation of targeted therapeutic and preventive strategies, thereby improving the quality of life for women affected by PD.McGill exercises, including the flexion test, side plank, and extension test, are widely used as assessment tests for the endurance and stability of the abdominopelvic region[15]. These exercises allow for the measurement of trunk muscle capacity to maintain specific positions against gravity, providing an objective assessment of muscular endurance[16].

Electromyography (EMG) is a valuable tool for assessing muscle activation and motor control,, as it provides detailed information about the pattern of muscle activation and its distribution[17]. Using EMG, practitioners can tailor exercise programs to optimize muscle engagement and address any imbalances or weaknesses[18,19]. In the context of PD, EMG can help investigate how pain and dysfunction affect muscle activation patterns, especially in the abdominopelvic region. By quantifying the electrical activity of muscles, EMG can reveal imbalances between agonist and antagonist muscle activity and shed light on the relationship between pain perception and motor control ability. This information is essential for developing effective physiotherapeutic interventions tailored to the needs of women with dysmenorrhea.

The aim of the present study was to explore the pattern of muscle activation with EMG during McGill stabilization exercises and to evaluate the endurance capacity of the abdominolumbopelvic musculature by comparing women with PD and those without this condition. Additionally, as a secondary objective, the study examined the influence of fatigue and compared the distribution of muscle activation at the beginning and end of the exercise.The underlying hypothesis of this study was that women experiencing PD may have alterations in motor control and decreased muscular endurance compared to healthy women.

Materials and methods

Study design

A single-center, analyst-blinded, observational, cross-sectional study was conducted at the Universidad Europea de Madrid, Spain (UEM) from September 2022 to March 2023. The sample consisted of an equal number of women with primary dysmenorrhea and women without dysmenorrhea, recruited among students and workers at UEM through informative posters distributed across all university buildings. Inclusion criteria were as follows: women diagnosed with PD by a physician from the medical service of UEM, over 18 years old, of reproductive age, and nulliparous. Exclusion criteria included the presence of any previously diagnosed osteoarticular disorder, use of hormonal contraceptives, the use of intrauterine devices (IUDs) and diagnosis of endometriosis or any other urogynaecological pathology. Participants in the control group had to meet the same requirements as those in the dysmenorrhea group, except for the absence of PD.

The local Research and Ethics Committee of UEM approved the study protocol with code CIPI/23.146. The study was conducted in accordance with the principles outlined in the Declaration of Helsinki, and all participants provided written informed consent. Voluntary subjects visited the study center on Day 0 to assess eligibility criteria and provide demographic data, including the day of their menstrual cycle, until the required sample size was attained. Study codes were used for participant identification, ensuring blinding of the analyst during subsequent electromyographic record analysis.

Trunk musculature endurance

To evaluate the endurance of the muscles in the trunk, isometric trunk tests as outlined by McGill et al. were conducted[15]. The four McGill tests: flexion, extension, and side plank (each side), have demonstrated outstanding reliability upon repetition[20]. The rest period between tests was 2 min to avoid muscle fatigue between tests (Fig. 1).

Mc Gill 1: trunk flexor endurance test

To assess the strength of the trunk flexor muscles, the flexion test was performed. Participants were seated on a bench, leaning their trunk against a support inclined at 60° to the horizontal. The knees and hips were flexed to 90° and the arms were crossed over the chest with the hands placed on the opposite shoulder. Participants were instructed to maintain this position while the support wedge was removed to start the test. Resistance time was recorded in seconds with a stopwatch, starting from the moment the participant's body stopped touching the wedge until their upper body dropped below the 60° angle.

Mc Gill 2 and 3: side plank test

The bilateral lateral plank test was performed to assess the strength of the lateral trunk muscles. Participants laid in a lateral decubitus position on an exercise mat with legs extended. The upper foot was placed in front of the lower foot for support. Participants were instructed to lift their hips off the mat to maintain a straight line across the body, supported by one elbow and the feet. The arm of the non-tested side was placed on the chest with the

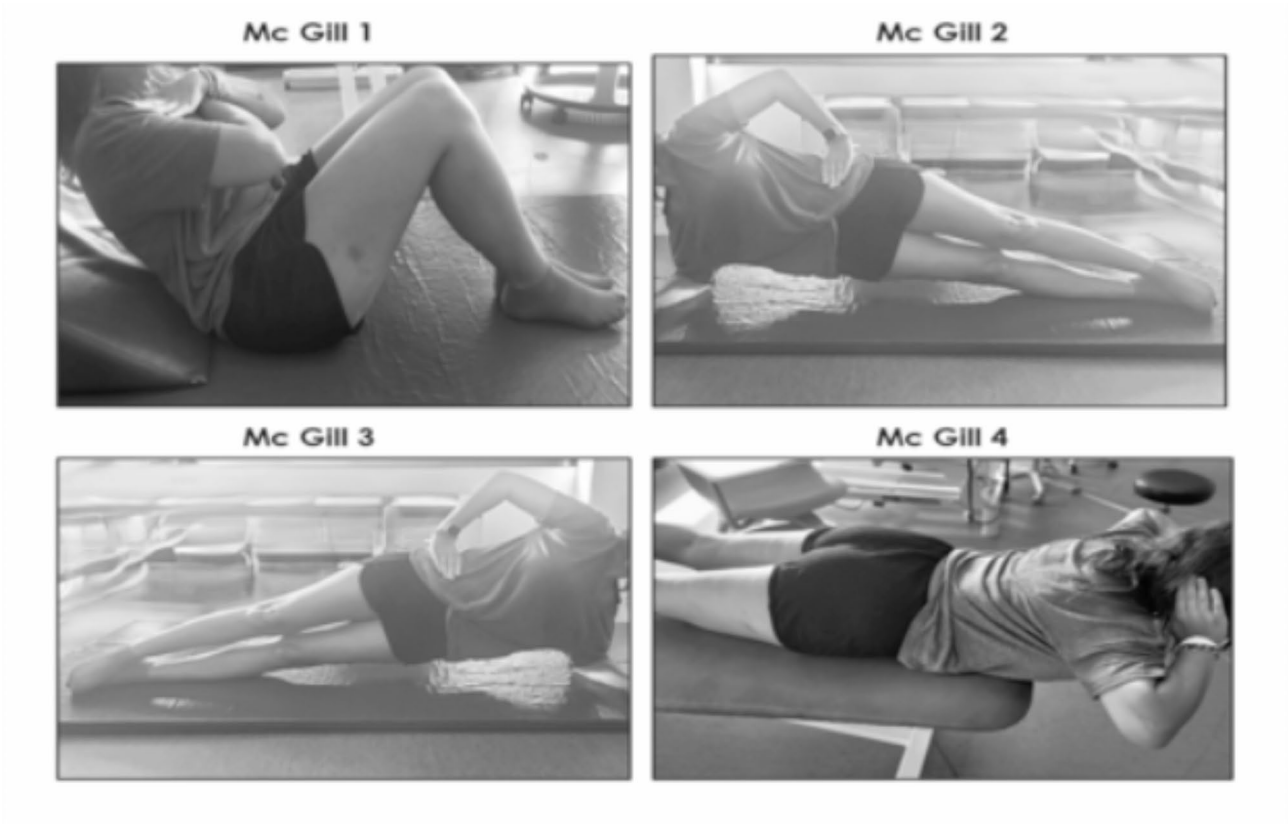

Fig. 1. McGill tests.

hand on the opposite shoulder. Endurance time was recorded in seconds with a stopwatch, and the test ended when the hip returned to contact with the mat. Electromyographic electrodes were positioned on the right side, thus recording muscle activity from the inferolateral side on the right plank and the supralateral side on the left plank.

Mc Gill 4: Extensor trunk endurance test
To assess the musculature of the posterior trunk, the extension test was conducted. Participants were placed in a prone position with their lower limbs securely fastened to the bench by the physiotherapist, and the trunk was positioned over the edge of the bench. At the beginning of the test, the upper limbs were crossed over the chest with hands resting on the opposite shoulders, and participants were instructed to lift their chests off the bench. They were then asked to maintain trunk extension for as long as possible. The duration of the exercise was recorded in seconds using a stopwatch, starting from the moment the participant assumed the horizontal position until their upper body made contact with the ground again or until 2 min had passed.

EMG procedure

Electromyographic activity during McGill's test was then measured: the first 10 s at the beginning of the exercise and the last 10 s before the exercise ended.

These exercises included detailed movements targeting specific muscle groups such as the external oblique (EO), rectus abdominis (RA), lumbar erector (LE), and gluteus medium (GM). To facilitate electromyographic recordings, the validated surface EMG mDurance® system (mDurance Solutions SL, Granada, Spain) was utilized[21]. This system comprised three components: sensors (Shimmer sensor, Shimmer Research Ltd., Dublin, Ireland), mobile computing Galaxy A7 Android Tablet (ZtotopCase, Suwon, Republic of Korea), and cloud-based data analysis.

For electromyographic activity recording, electrodes were strategically placed on specific anatomical landmarks to target the EO, RA, LE, and GM muscles (Table 1)[22–24]. Electrodes used were pre-gelled Ag/AgCl (Ref. 019-400400, Natus Medical Incorporated, Middleton, WI, USA) and positioned unilaterally on the right side of the body (Fig. 2).

The first sensor (MDUR-4B1A) had wires connected to electrodes placed on the EO and RA. An additional electrode was placed on the ASIS as a reference electrode. The sensor was then strapped to the front of the leg. The second sensor (MDUR-4B05) was fitted with electrode cables to record the activity of the LE and GM. An electrode was also placed on the sacrum as a reference electrode. The sensor was then strapped to the dorsal spine.

Muscle	Electrode placement
EO	Two contiguous electrodes were placed in a craniocaudal direction at the triangle formed by the inguinal ligament, the anterosuperior iliac spine (ASIS) and the umbilical midline
RA	Two contiguous electrodes were placed longitudinally along the muscle belly at the level of the umbilicus
LE	Two contiguous electrodes were placed in the craniocaudal direction with 3 cm between them, lateral to the spinous process of L1
GM	Two electrodes were placed 2 cm apart lateral to the median sacral crest in the cranio-caudal direction

Table 1. Description of the electrode placement.

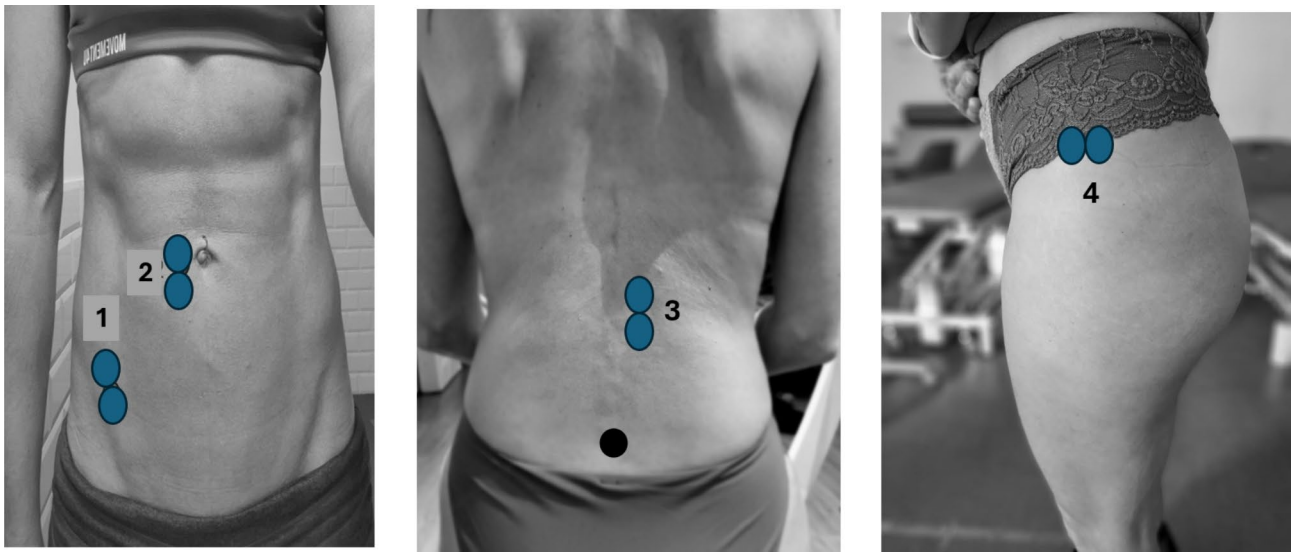

Fig. 2. Electrode placement: 1:EO, 2 RA, 3 LE, 4 GM.

EMG measurements were conducted in a controlled environment at 22 ± 1 °C and 40–60% humidity to minimize variability in skin conductivity. The laboratory was equipped with electromagnetic shielding and followed strict isolation protocols, including a common grounding system and 50/60 Hz filters to reduce electrical interference. Participants maintained stable, ergonomic positions, and supports were used to limit involuntary muscle activation. The skin was cleaned with isopropyl alcohol for better electrode adherence.

Signal normalization

The intervention started with the initial assessment of participants' electromyographic activity during a standing resting position for 2 s, serving as a baseline for subsequent analyses. Subsequently, participants underwent the maximum voluntary isometric contraction (MVIC) protocol for each muscle group. The MVIC assessment involved 3 s isometric contractions with 20 s of rest between repetitions.

For the MVIC assessment of the EO muscle, subjects were placed in a supine position on the table with the lower limbs flexed and were asked to perform an abdominal crunch with contralateral rotation. For the MVIC assessment of RA, subjects were placed in a supine position on the table with the lower limbs flexed and were asked to perform an abdominal crunch. For the MVIC assessment of GM, participants were placed in a lateral decubitus position on the table with the right leg flexed, and they were asked to perform a hip abduction. Finally, for the MVIC assessment of LE, participants were placed in a prone position on the table with the lower legs flexed, executing trunk extension.

EMG data analysis

In order to avoid potential interactions with the electrodes in our study, we have utilized the SENIAM protocol (Surface ElectroMyoGraphy for the Non-Invasive Assessment of Muscles)[25].

A high-pass filter with a frequency cutoff of 20 Hz was implemented to eliminate low-frequency noise and interference, while a low-pass filter with a frequency cutoff of 450 Hz was utilized to diminish high-frequency noise. These filter parameters were carefully chosen to capture the pertinent frequencies associated with muscle activity during a dynamic task. Subsequent to this filtering process, electromyographic signals were normalized using the MVIC values recorded at the initiation of the session. Following normalization, a comprehensive analysis of the normalized electromyographic data was undertaken, encompassing the determination of the total average, and maximum peaks during each exercise from the McGill protocol.

The rectification of the electromyographic signal was executed using mDurance® software version 1, incorporating absolute rectification to convert the original signal into a unipolar representation. This conversion

entailed taking the absolute value of each data point in the signal. The calculation of the "average of maximum peaks" involved averaging the first 10 peak values within each electromyographic analysis.

Normalization of the rectified signal was performed by computing the root mean square (RMS) over a specified interval. Subsequently, the rectified signal was divided by the RMS to express the amplitude in relative terms.

For each McGill exercise, the average activity of the four measured muscles was recorded over the total duration of the exercise and expressed as a percentage (Fig. 3). Additionally, the peak maximum activations during the entire exercise were also measured. Secondarily, we analyzed the average muscle activity during the first 10 s and the last 10 s of the exercise to observe if fatigue had any influence on both groups (PD and control).

Sample size calculation

The sample size calculation was performed with the GPower Software, using an alpha error of 0.05, a beta error of 0.2 and a low effect size (f = 0.23 or eta partial square of 0.05). Due to the study design, no estimated dropout rate was considered. Consequently, the determined total sample size was at least 40 participants, divided equally into two groups ($n = 20$ per group).

Statistical analysis

Statistical analyses were performed using SPSS version 29 for Windows (IBM, Armonk, NY, USA). The distribution of the data was assessed using the Shapiro–Wilk test for normality, and histograms were examined for visual confirmation. Data were reported as mean and standard deviation. Independent t-tests were used to compare the groups at baseline. Homoscedasticity and sphericity were verified. When these assumptions were met, a two-way analysis of variance (ANOVA) 2 (groups) × 2 (times) was conducted. The effect size was estimated using partial eta squared (η^2), with values interpreted as 0.01 (small), 0.06 (medium), and 0.14 (large). The U Mann–Whitney test was used to compare percentages of average activity and of peak values of EMG activation of the four measured muscles between study groups. Leves of significance was set at ≤ 0.05.

Results

A total of 44 female participants were included in the study ($n = 22$ for the dysmenorrhea group and $n = 22$ for the control group). The total sample had a mean age of 26.16 ± 6.97 years, a mean weight of 61.24 ± 9.88 kg, mean height of 164.91 ± 6.46 cm and a mean BMI of 22.44 ± 2.97 kg/m^2. The dysmenorrhea group and control

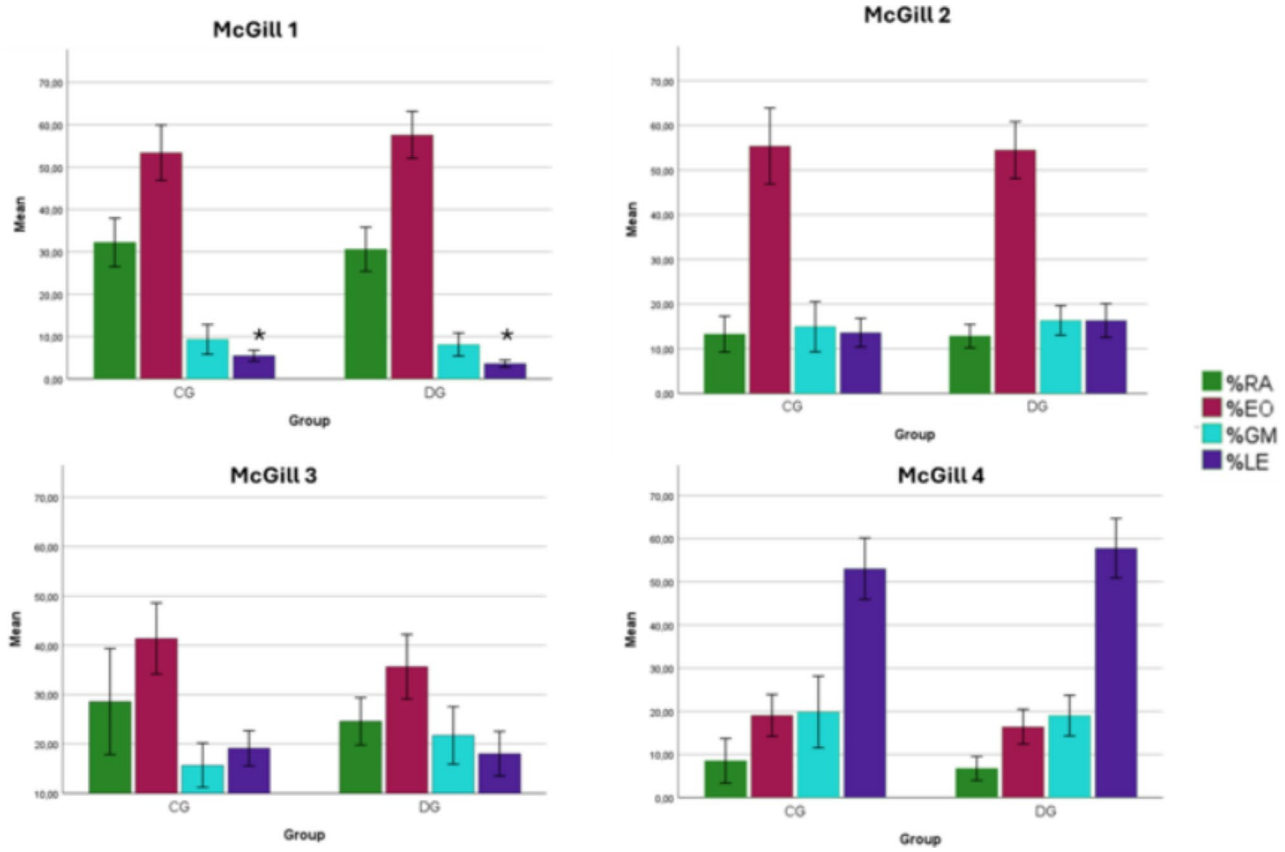

Fig. 3. Average activation (expressed as percentage) of each muscle over the total duration of each of the McGill tests. * = statistically significant ($p = 0.016$). *RA* rectus abdominis, *EO* external oblique, *GM* gluteus medius, *LE* lumbar erector.

group had, respectively, a mean age of 27.18 ± 6.51 vs. 25.14 ± 7.41 years, a mean weight of 61.32 ± 9.02 vs. 61.16 ± 10.89 kg, height of 166.00 ± 5.98 vs. 163.82 ± 6.86 cm, and a mean BMI of 22.18 ± 2.64 and 22.70 ± 3.31 kg/m^2. No significant differences were observed between groups for age ($p = 0.337$), weight ($p = 0.958$), height ($p = 0.267$) and BMI ($p = 0.567$).

Participants were evaluated around day 15 of the menstrual cycle on average (15.30 ± 7.40 days). In the control group, the mean was 14.05 ± 7.38 days, while in the dysmenorrhea group it was 16.55 ± 7.39 days. The difference between the groups was not statistically significant ($p = 0.254$), indicating that both groups were at a similar stage in the menstrual cycle at the time of evaluation.

Table 2 shows the trunk endurance times for the McGill test. No significant differences were observed between groups for any of the measurements ($p > 0.05$) (Table 2).

For the Mc Gill 1, the analysis of the RA and EO muscles showed no significant interaction effect between time and group. The F values were 0.014 and 0.105, with p-values of 0.906 and 0.748, and partial eta-squared (η^2p) of 0.000 and 0.003, respectively (Fig. 3). For the Mc Gill 2:, the RA, EO, LE, and GM muscles also showed no significant effects, with F values ranging from 0.074 to 0.380, p-values from 0.523 to 0.787, and η^2p between 0.002 and 0.010 (Fig. 4). For the Mc Gill 3 continued this trend with non-significant results across RA, EO, LE, and GM muscles, with F values between 0.271 and 1.372, p-values from 0.248 to 0.605, and η^2p from 0.007 to 0.033 (Fig. 5). Similarly, the Mc Gill 4 showed no significant interaction effects in RA, EO, and LE muscles, with F values of 0.402 to 1.846, p-values of 0.182 to 0.529, and η^2p ranging from 0.010 to 0.044 (Fig. 6).

Table 3 shows the peak activity of each muscle during the McGill tests. A significant difference between groups was only found for the LE activation during the McGill 1 test.

Figure 3 shows the average activation (expressed as percentage) of each muscle over the total duration of each of the McGill tests. Again, a significant difference between groups was only found for the LE activation during the McGill 1 test.

Figures 4, 5, 6 and 7 show the peak electromyographic activity of the RA, EO, LE, and GM muscles during the McGill trunk endurance tests at two time points: the initial 10 s at the start of the exercise and the final phase (last 10 s of the exercise). There were no significant differences between groups during the initial and terminal phase of any of the McGill trunk endurance tests ($p > 0.05$).

Discussion

This study explored the muscle activation pattern using surface EMG during McGill stabilization exercises and the endurance capacity of the abdominolumbopelvic musculature, comparing women with PD to those without this condition.

Our results revealed significant differences in the activation of the LE muscle during the "McGill 1: flexor endurance test." Both the mean activation and peak values were lower in women with dysmenorrhea compared to the control group. However, no significant differences in EMG activation were observed in the RA, OE, or GM during this same exercise, nor in any of these muscles during the "McGill 2 and 3: side plank" or the "McGill 4: trunk extensor endurance test."

The reduced activation of the LE, an antagonist to the trunk flexors, in women with dysmenorrhea suggests a possible inhibition of the antagonist musculature. This could be a compensatory response to recurrent pain associated with PD, aimed at minimizing stress on painful areas during movement or exercise. Previous studies, such as those by Iacovides et al.[26] and Karakus et al.[27], have linked dysmenorrhea to reduced physical performance, although they did not focus specifically on the relationship between agonist and antagonist muscle activity. Iacovides et al.[26] reported that severe dysmenorrhea negatively affects quality of life and daily performance, but did not specifically assess muscle capacity. In contrast, Karakus et al.[27] suggested that dysmenorrhea may decrease endurance in certain exercises due to associated pain and discomfort, although they did not directly assess EMG activity.

In addition, the significant reduction in the electromyographic activation of the LE during the McGill 1 test suggests a specific muscular inhibition in the context of an abdominal crunch exercise, potentially increasing the risk of injury due to reduced active spinal protection[28]. This observation underscores the need for further studies to evaluate the behavior of the abdomino-lumbar muscles during other movements and physical efforts, particularly those in which the spinal erectors function as antagonists, to determine whether the observed motor control changes persist under dynamic conditions.

Current literature suggests that increased muscle activation may serve as a load-buffering mechanism, which is associated with structural adaptations such as an increase in muscle cross-sectional area[29], thereby contributing to improved joint stability[30]. Conversely, reduced activation, especially under peak demand conditions, has been linked to latency responses that may hinder adequate muscular contraction in instances of elevated intra-abdominal pressure, such as during abdominal exercises, potentially raising the risk of urinary incontinence[31].

Variables	Control group ($n = 22$)	Dysmenorrhea group ($n = 22$)	p	Mean difference (95% CI)
Mc Gill 1: Trunk flexor	76.81 ± 30.67	69.81 ± 34.03	0.478	6.99682 (− 12.71; 26.71)
Mc Gill 2: Right side plank (s)	52.31 ± 24.16	47.90 ± 23.10	0.478	4.40909 (− 9.97; 18.79)
Mc Gill 3: Left side plank (s)	50.19 ± 19.03	45.34 ± 16.83	0.539	4.85227 (− 6.08; 15.78)
Mc Gill 4: Extensor trunk (s)	80.83 ± 33.36	81.17 ± 34.89	0.974	− 0.34091 (− 21.11; 20.43)

Table 2. Trunk endurance times (s) for the McGill tests. *s* seconds. Data are expressed as mean ± SD.

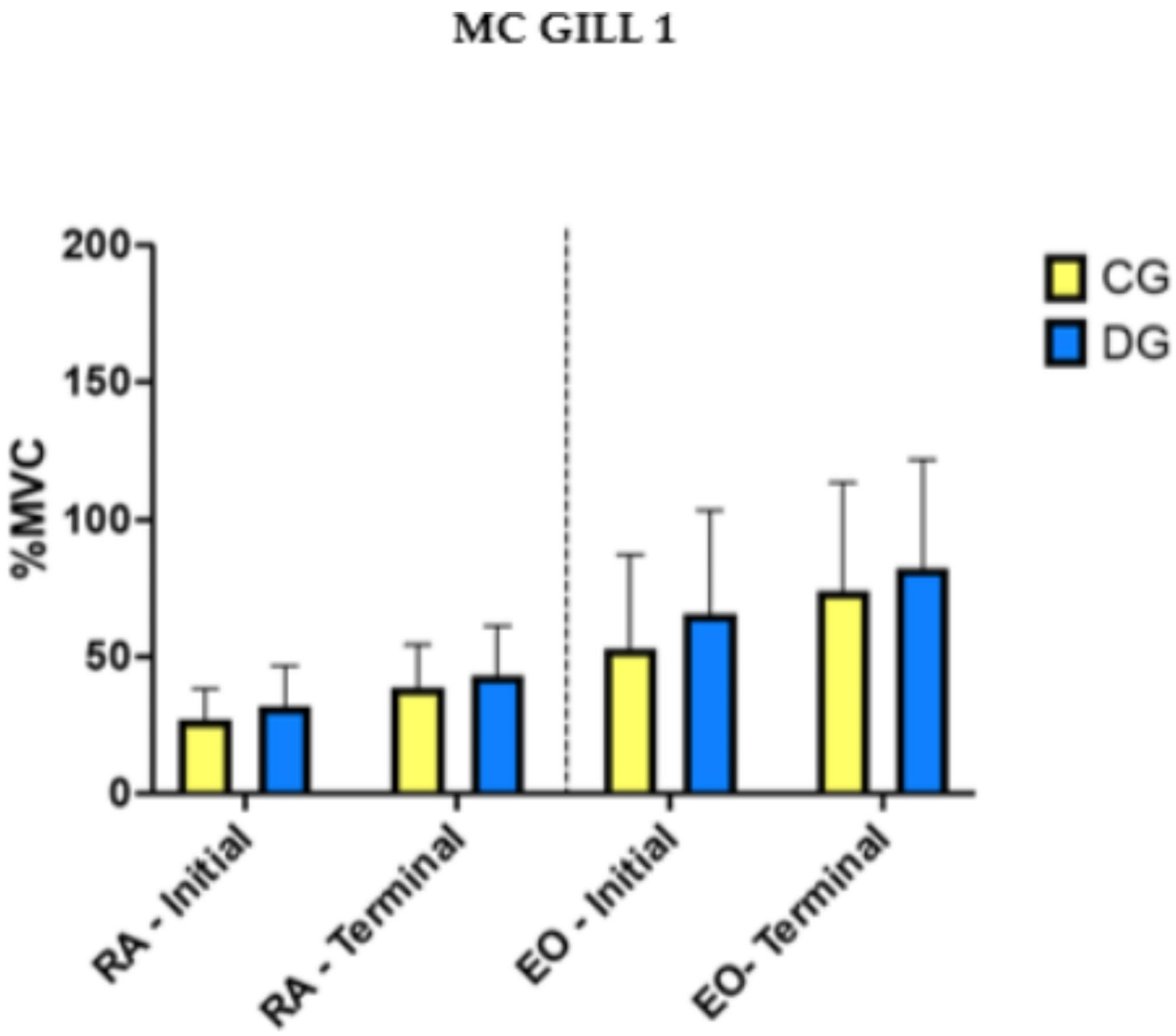

Fig. 4. Mean electromyographic activity during the Mc Gill 1: flexor trunk endurance test.

Dynamic stability of the pelvic cavity is essential not only for physical performance but also for dysfunction prevention, relying on the synergistic and precise activation of the abdominal, gluteal, and other related muscle groups[32]. Optimal coordination, regulated by the central nervous system, is necessary to maintain stable intra-abdominal pressure and uniform load distribution; any delay in muscle activation could compromise lumbo-pelvic stability and, consequently, increase the risk of injury[33].Our findings offer new insights by suggesting that the pain associated with dysmenorrhea may selectively impact the activation of specific muscles, potentially contributing to muscle coordination imbalances during trunk endurance tasks. This highlights the need for further research into antagonist muscle inhibition in women with PD and its possible impact on motor control and functionality.The lack of significant differences in muscle activation in other exercises and in the RA, OE, or GM raises questions about a possible dissociation between perceived pain and functional capacity in controlled settings. The absence of differences in these tests could be due to compensatory adaptations allowing these women to maintain physical function during pain-free phases of the menstrual cycle. Such adaptations have been reported in other chronic pain conditions, where individuals often develop coping strategies to preserve functionality despite their condition[34]. For instance, people with chronic pain frequently use psychological resilience and cognitive-behavioral strategies to continue with daily activities and maintain a degree of normalcy[35].

These adaptive mechanisms, including psychological resilience, may explain the absence of changes in muscle activation. Furthermore, individual variability in muscle response, pain threshold, and pain tolerance likely plays a crucial role in physical performance. Psychological factors and quality of life should also be considered, as they significantly influence outcomes in patients with dysmenorrhea[36]. The presence or absence of central sensitization in women with PD remains controversial. Lee et al.[37] reported that while pain perception alters during the painful phase of the menstrual cycle, these changes do not persist throughout the cycle. This may

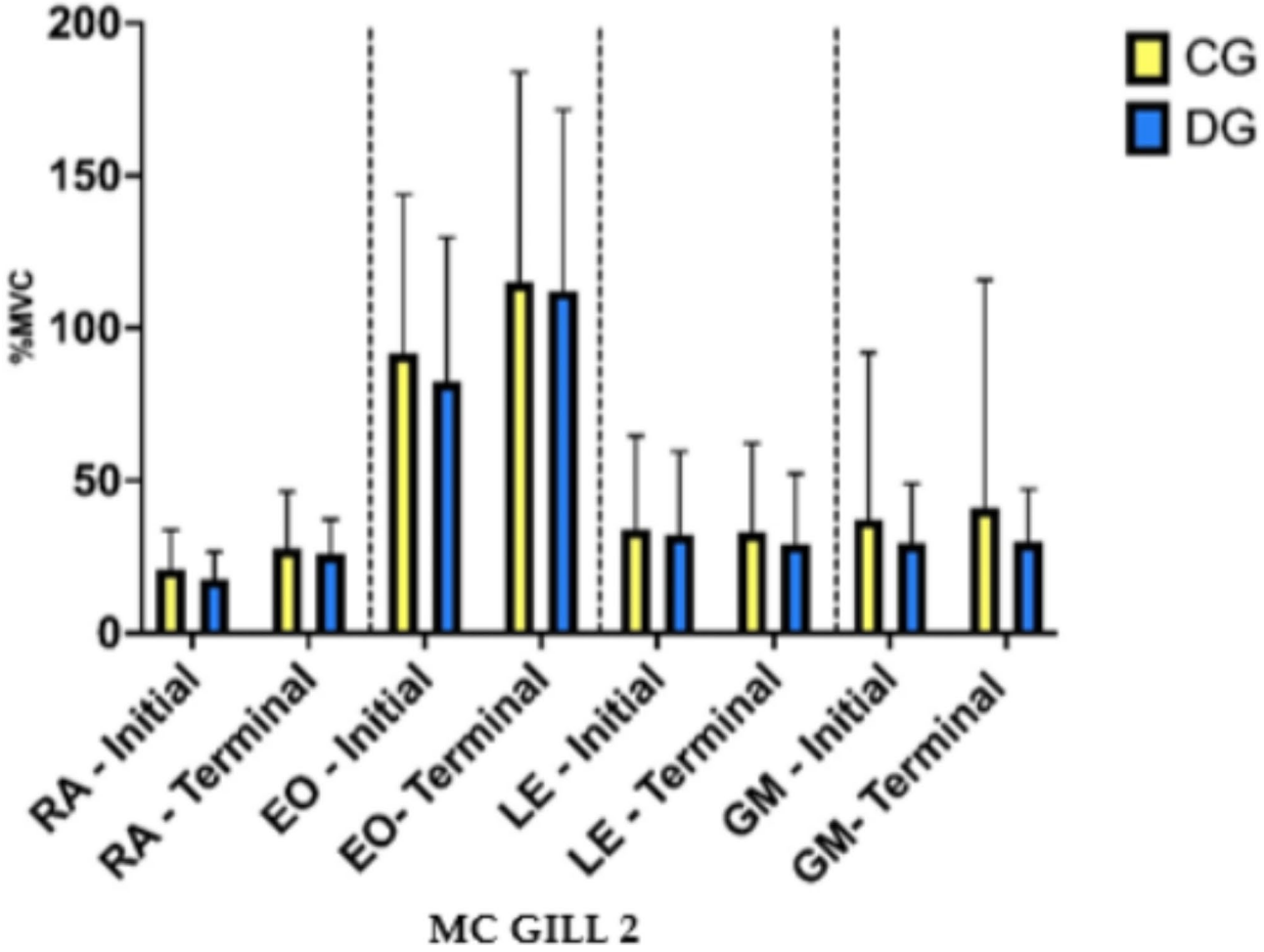

Fig. 5. Mean electromyographic activity during the Mc Gill 2: right side plank.

explain why motor control modifications are associated with the painful period but are absent during the rest of the cycle.

Another important finding of our study was the lack of significant differences in endurance, meaning the time the exercises were sustained between the groups. In line with our findings, Potur et al.[38] found that pain associated with dysmenorrhea did not significantly reduce muscle strength but had a considerable impact on motivation to engage in physical activities. These results emphasize the importance of addressing both physical and psychological factors when assessing the effects of dysmenorrhea on physical performance, as previously suggested[39].

Conversely, Özdemir et al.[40] found that performance did not decrease in athletes without dysmenorrhea, but endurance was negatively affected in athletes with PD. The discrepancy with our results may stem from the fact that Özdemir's study focused on athletes and assessed performance during painful phases of the menstrual cycle. Similarly, Karakus et al.[27] reported that trunk muscle endurance was significantly lower in women with PD compared to asymptomatic controls. This difference may be due to the timing of their evaluations, conducted specifically on the last day of menstruation, and the participants' level of physical activity.

Additionally, Lebrun et al.[41] found that the menstrual cycle phase did not significantly affect performance indices like aerobic capacity, anaerobic capacity, isokinetic strength, and high-intensity endurance, although it was unclear how many participants had dysmenorrhea. Likewise, other studies suggest that physical performance is not significantly affected by the menstrual period and that pain decreases during training and competition in athletes[42]. This could indicate that, despite experiencing pain during specific days of the menstrual cycle, women are capable of maintaining comparable physical function during pain-free phases.

In conclusion, while dysmenorrhea is associated with significant pelvic pain, our study suggests that pain does not necessarily directly impact muscle endurance.

Regarding muscle fatigue, no significant differences were found in any muscles. The absence of changes in both superficial and deep muscle activation throughout the exercises indicates no loss of motor control due to fatigue. Therefore, the pain experienced during the menstrual cycle in women with dysmenorrhea may not directly affect basic muscle capacities during other phases of the cycle.

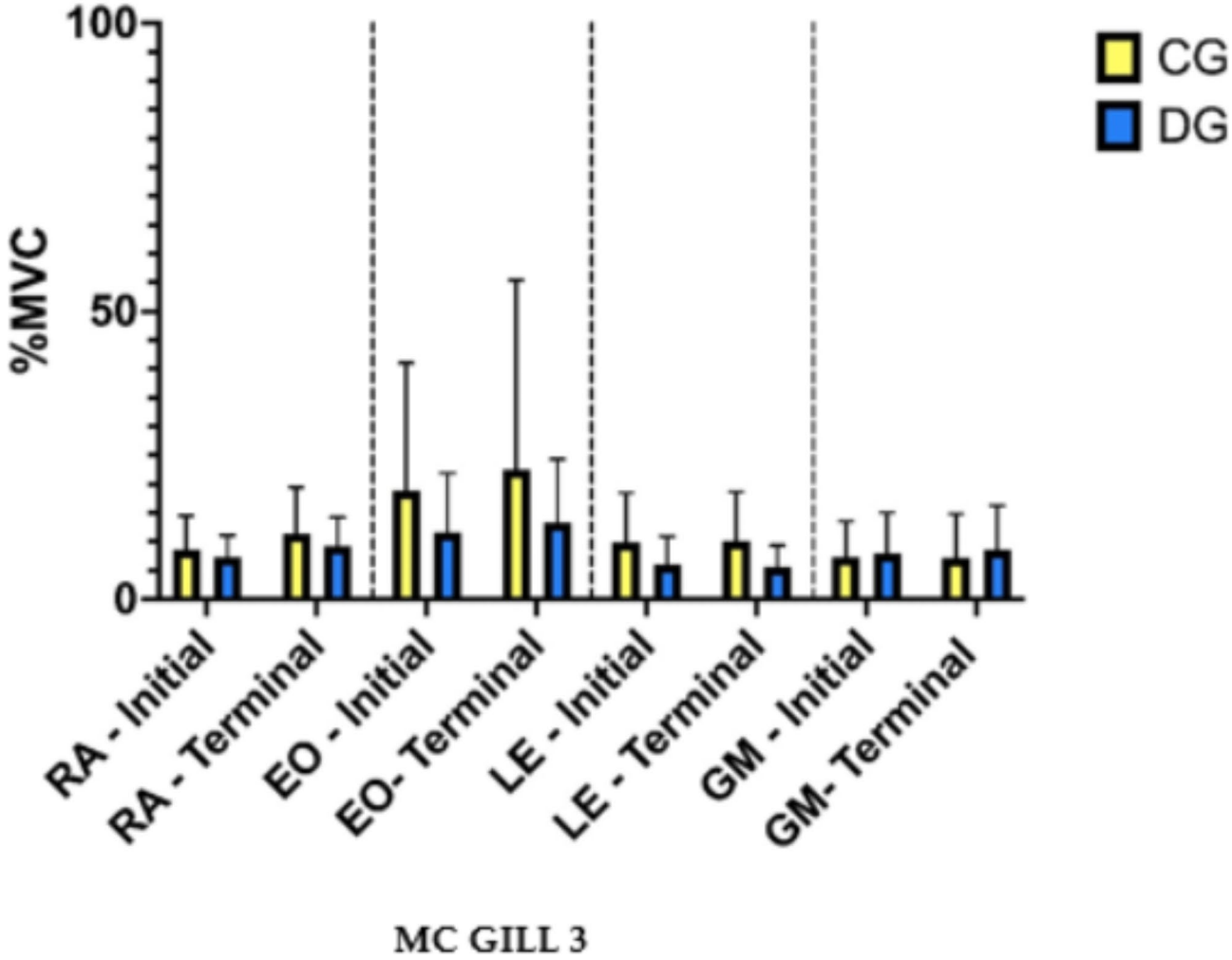

Fig. 6. Mean electromyographic activity during the Mc Gill 3: left side plank.

Adaptation and acceptance of pain may also play a crucial role in the quality of life of women with dysmenorrhea. McCracken et al.[43] confirmed that better pain coping strategies are associated with less disruption of motor patterns in chronic pain conditions. Although our results do not show motor control alterations, a comprehensive approach addressing pain adaptation and acceptance could prevent future impairments and improve the quality of life and performance of women with PD.

Limitations

Our study shares common limitations with previous research in this area, such as hormonal variability. In addition, the phase of the menstrual cycle during which the tests were performed (approximately in the middle of the luteal phase) could have significantly affected the results. The small sample size is another limitation, which could reduce the generalisability of our results. In addition, factors such as individual pain tolerance, psychological influences on pain perception and lifestyle variables were not fully controlled for. These elements should be carefully addressed in future studies to provide a more accurate and comprehensive assessment of the relationship between dysmenorrhoea and physiological responses.

Future lines of research

Although our study provides valuable information on muscle function and pain response in people with PD, the results must be interpreted within the context of the study, particularly due to the limited sample size and the single-phase assessment of the menstrual cycle. To improve generalisability, future studies should aim to assess muscle function in diverse populations and settings, including cross-cultural studies and varied age groups. Longitudinal assessments throughout the menstrual cycle would also allow for a more nuanced understanding of hormonal and cycle-dependent variations in muscle function and pain perception.

The presence of potential confounding factors, including participants' exercise history and severity of dysmenorrhoea symptoms, may significantly influence the results of the study.

In addition, interventions that integrate pain management, physiotherapy and psychological support are worth investigating, as a multidisciplinary approach may offer more robust relief and address the multifaceted

Variables	Control group ($n = 22$)	Dysmenorrhea group ($n = 21$)	*P* value
McGill 1			
RA	93.9 (53.4, 127.0)	87.8 (63.9, 100.7)	0.662
EO	162.0 (96.2, 189.6)	182.19 (111.4, 227.0)	0.285
GM	20.2 (12.2, 26.6)	20.2 (10.1, 30.2)	0.942
LE	14.4 (9.6, 18.7)	10.6 (6.8, 13.2)	0.049
McGill 2			
RA	62.3 (39.6, 69.8)	59.0 (37.7, 78.1)	0.923
EO	266.6 (188.4, 473.8)	223.47 (129.3, 321.4)	0.126
GM	58.8 (37.0, 111,1)	81.1 (48.0, 112,3)	0.644
LE	54.1 (37.0, 81.8)	75.3 (42.9, 103.0)	0.290
McGill 3			
RA	24.9 (16.6, 44.8)	23.2 (16.9, 33.4)	0.264
EO	82.0 (48.1, 132.7)	63.5 (40.9, 102.1)	0.138
GM	23.7 (17.1, 38.8)	21.9 (13.4, 56.5)	0.697
LE	33.1 (11.9, 51.7)	15.1 (9.2, 30.0)	0.109
McGill 4			
RA	13.8 (5.3, 49.4)	7.0 (4.0, 15.6)	0.065
EO	33.9 (23.2, 81.6)	21.4 (14.7, 37.1)	0.065
GM	21.4 (17.8, 40.5)	21.6 (10.3, 31.1)	0.437
LE	77.9 (63.7, 99.1)	76.8 (62.1, 93.0)	0.734

Table 3. Peak values of the electromyographic activity of the different muscles during the McGill exercises in both study groups. *RA* rectus abdominis, *EO* external oblique, *GM* gluteus medius, *LE* lumbar erector. Data are expressed as median (P25, P75).

effects of dysmenorrhoea. In this sense, recent studies have documented the effectiveness of physiotherapeutic interventions for alleviating symptoms associated with PD. One study demonstrated that manual therapy and pelvic floor exercises, both separately and in combination, significantly improved pain and quality of life in women with dysmenorrhea[44]. Another complementary study found that a bilateral global pelvic manipulation technique reduced lumbopelvic pain and improved pain thresholds in the sacroiliac joints, along with serotonin levels[45]. These findings emphasize the importance of physiotherapeutic interventions for addressing pain associated with PD and underscore the need to investigate the impact of these techniques on muscle activation and coordination, which could contribute to enhanced functionality and quality of life for affected women. Expanding these lines of research would improve the applicability of the findings and support the development of specific and effective interventions for this population.

Conclusions

No differences were found in the muscle activation of the abdominolumbopelvic region or in endurance in the McGill exercises between women with and without dysmenorrhea. This study reflects a complex interaction between pain, muscle function, and physiological adaptations, highlighting the importance of a multifaceted and well-contextualized approach to understanding and treating dysmenorrhea.

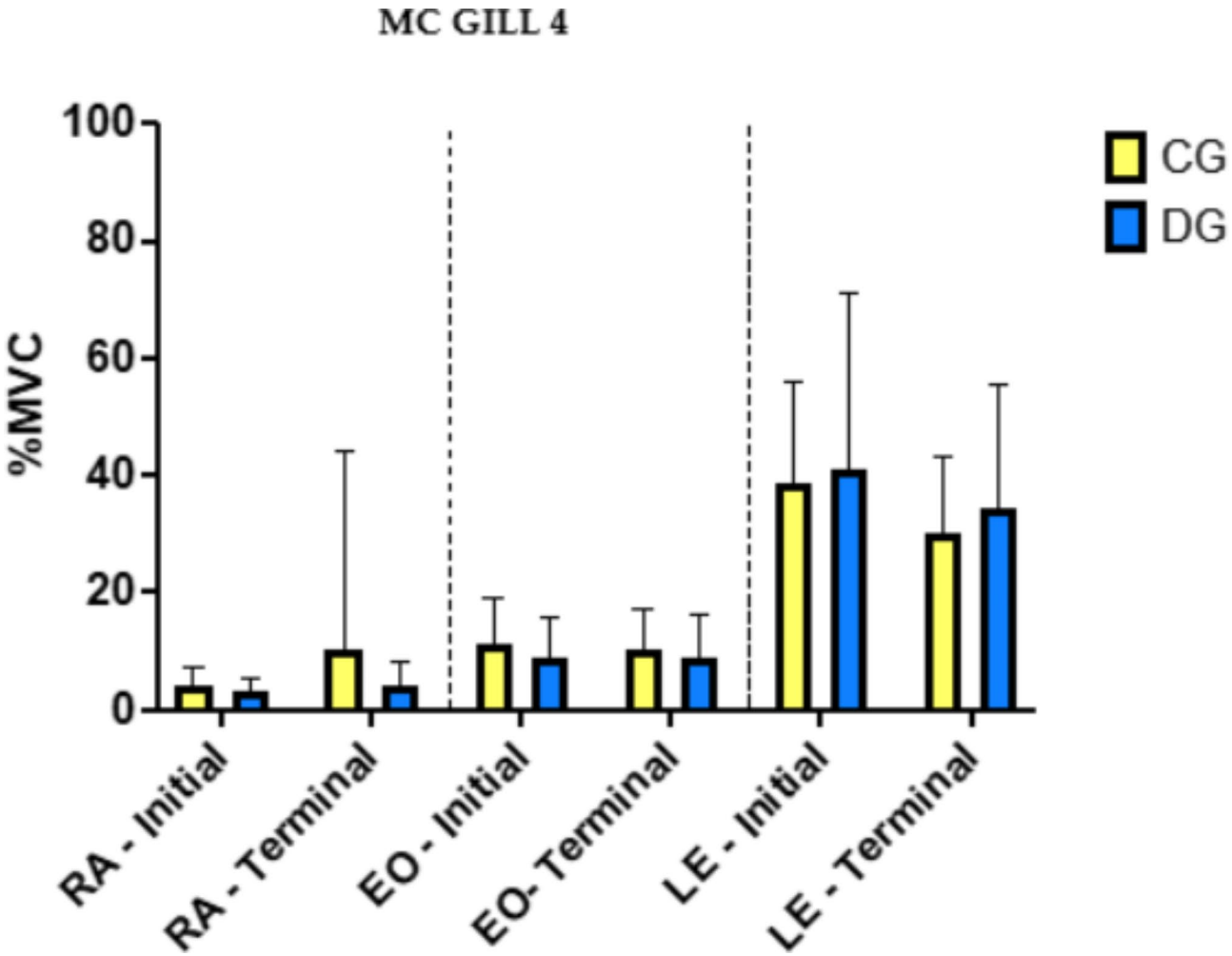

Fig. 7. Mean electromyographic activity during the Mc Gill 4: extensor trunk endurance test.

Data availability

Datasets supporting the reported results are available upon request to the first author.

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Author contributions

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Declarations

Competing interests
The authors declare no competing interests.

Institutional Review Board Statement
The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Commission of Universidad Europea de Madrid (protocol code CIPI/23.146; date of approval: April 20, 2023).

Informed consent
Informed consent was obtained from all participants involved in the study.

Additional information
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